

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	4 Marks

Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Home, Predict, Result, About, Applicant Profile and History pages	HTML5, CSS3, JavaScript, Font Awesome
2	Application Logic	Handles form submission, preprocessing and routing	Python (Flask)
3	Data Preprocessing	Missing-value handling, label encoding, feature scaling	Pandas, NumPy, Scikit-learn
4	Machine Learning Models	Model training and comparison	Decision Tree, Random Forest, KNN, XGBoost (Scikit-learn, XGBoost)
5	Deployed Model	Final prediction model in production	K-Nearest Neighbors (KNN), 84.55% accuracy
6	Model Persistence	Saved trained artifacts	Joblib – model.pkl, encoder.pkl, scaler.pkl
7	Visualization	Exploratory data analysis and model comparison charts	Matplotlib, Seaborn
8	Version Control	Source code hosting	GitHub

Repository & Deliverables

- Source Code (GitHub): https://github.com/anushaobulapuram/smart_lender_smart_bridge
- Project Drive Link:
<https://drive.google.com/file/d/1SFsukLL6qTV93eniIDsgYuXs0K5jTuJl/view?usp=drivesdk>